Title of Project

by

First Author
243014001

Capstone Project (CSE 400) submitted in partial fulfillment of the requirements for the degree of

Bachelor of Science in Computer Science and Engineering

Under the supervision of

Supervisor's Name

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GREEN UNIVERSITY OF BANGLADESH

FALL 2025

DECLARATION

Title	Title of Project
Author	*First Author*
Student ID	243014001
Supervisor	Supervisor's Name

I declare that this project entitled *Title of Project* is the result of my own work except as cited in the references. The project has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

First Author
243014001

Department of Computer Science and Engineering
Green University of Bangladesh

Date: November 24, 2025

CERTIFICATE OF ENDORSEMENT

This is to certify that the project entitled *Title of Project* by **First Author** (Student ID: 243014001) has been successfully defended and examined.

Based on the evaluation and discussion held on **November 24, 2025**, we, the undersigned, hereby endorse the project for acceptance and approval.

External Member's Name
External Member, Board: 10

External Member's Position
External Member's Affiliation

Board Member's Name
Board Member, Board: 10

Board Member's Position
Department of Computer Science and Engineering
Green University of Bangladesh

Board Chair's Name
Chair, Board: 10

Board Chair's Position
Department of Computer Science and Engineering
Green University of Bangladesh

Date: November 24, 2025

GREEN UNIVERSITY OF BANGLADESH

Department of Computer Science and Engineering
Green University of Bangladesh
Kanchon, Rupganj, Narayanganj - 1461

CERTIFICATE

This is to certify that the project entitled **Title of Project**, submitted by **First Author** (Student ID: 243014001) an undergraduate student of the **Department of Computer Science and Engineering** has been examined. Upon recommendation by the examination committee, we hereby accord our approval of it as the presented work and submitted report fulfill the requirements for its acceptance in partial fulfillment for the degree of *Bachelor of Science in Computer Science and Engineering*.

Supervisor's Name
Supervisor's Position

Department of Computer Science and Engineering
Green University of Bangladesh

Name of the Chairperson
Designation and Position

Department of Computer Science and Engineering
Green University of Bangladesh

Place: Narayanganj
Date: November 24, 2025

ACKNOWLEDGMENTS

I would like to express my deep and sincere gratitude to my research supervisor, *Supervisor's Name*, for giving me the opportunity to do research and providing invaluable guidance throughout this work. His dynamism, vision, sincerity and motivation have deeply inspired me. He has taught me the methodology to carry out the work and to present the works as clearly as possible. It was a great privilege and honor to work and study under his guidance.

I am greatly indebted to my honorable teachers of the Department of Computer Science and Engineering at the Green University of Bangladesh who taught me during the course of my study. Without any doubt, their teaching and guidance have completely transformed me to a person that I am today.

I am extremely thankful to my parents for their unconditional love, endless prayers, caring and immense sacrifices for educating and preparing me for my future. I would like to say thanks to my friends and relatives for their kind support and care.

Finally, I would like to thank all the people who have supported me to complete the project work directly or indirectly.

First Author
Green University of Bangladesh
Date: November 24, 2025

Dedicated to Almighty Allah —
All praise to Him for His endless mercy and guidance. Dedicated to my beloved parents for their love and support, and to all who inspired and encouraged me along the way.

– First Author

ABSTRACT

Summarize the rationale for your project, the methodology, anticipated results, and conclusions you will draw from your work. (roughly 300 words)

Keywords: keyword1, keyword2

Contents

List of Figures

List of Tables

1. Introduction

1.1 Background of the Project

Write the background information of your project here. Explain the context, importance, and relevance of the project topic.

1.2 Problem Statement

Describe the main problem your project aims to solve. Mention the existing issues, gaps, or inefficiencies that motivated the study.

1.3 Motivation

Discuss the reasons or inspirations behind selecting this project. You may include real-world needs, academic curiosity, or potential impact.

1.4 Project Objectives

1.4.1 General Objective

State the overall aim or goal of the project in broad terms.

1.4.2 Specific Objectives

List the specific, measurable, and achievable objectives that will help fulfill the general objective.

1.5 Project Scope

Define the boundaries of the project—what it will cover and what it will not.

1.6 Project Deliverables

List the expected outputs or results from the project (e.g., software, reports, documentation, models, etc.).

1.7 Financial Planning and Timeline

1.7.1 Budget Estimation

Instruction: Break down the costs of your work if it involves hardware, software, or travel. If costs are not applicable, skip this part or write: "No major cost is involved."

Example: No major costs are involved in this work. All tools and datasets used are open-source and freely available. The development environment runs on a personal laptop.

1.7.2 Gantt Chart

Instruction: "Create a visual chart showing your research timeline (tasks vs. weeks/months)".

Example: The Fig 1.1 represents a visual chart of the research timeline for a heart disease prediction project. It outlines the sequence of key tasks, including data collection, data pre-processing, feature selection, model development, training and validation, performance analysis, and deployment preparation. Each task is mapped along a timeline to show the planned duration and order of activities, helping to organize the workflow and track progress throughout the study.

1.8 Report Organization

Explain how the report is structured (e.g., what each chapter contains).

Stages of research	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7
Selection of topic							
Data collection from secondary sources							
Literature review							
Research methodology plan							
Selection of the Appropriate Research Techniques							
Analysis & Interpretation of Data							
Findings and recommendations							
Final research project							

Figure 1.1: A planned thesis schedule.

2. Literature Review

A literature review is a comprehensive examination and synthesis of existing scholarly works, including research articles, books, and other relevant sources, about a specific topic or research question. It plays a crucial role in academic writing by establishing the context of the research, identifying gaps and trends in the current knowledge landscape, and providing a theoretical framework for the study. Through a critical evaluation of the literature, it justifies the need for the proposed research, guides methodological choices, and helps avoid redundancy by building upon prior work. The literature review not only assesses the quality and credibility of existing research but also organizes information in a structured manner, categorizing studies based on themes or methodologies. Additionally, it serves as a roadmap for future research by highlighting unanswered questions and areas that warrant further investigation. Overall, a well-conducted literature review contributes to the depth and credibility of academic research, ensuring that the study is situated within the broader scholarly discourse.

2.1 Introduction to Related Works

Provide an overview of the related studies, projects, or research areas connected to your topic. Discuss how they contribute to understanding your problem domain.

2.2 Existing Systems / Solutions

Describe existing systems, tools, or methods that address similar problems. Highlight their features, strengths, and weaknesses.

2.3 Technology Background

Explain the key technologies, frameworks, programming languages, or hardware components relevant to your project. Include a brief discussion of why they were chosen or how they are used in existing systems.

2.4 Comparative Analysis of Existing Approaches

Compare and contrast various existing solutions. Discuss how they differ in methodology, performance, cost, scalability, or other important criteria.

2.5 Identified Gaps and Opportunities

Identify the shortcomings or gaps in existing systems or research. Explain how your project aims to address these gaps and what opportunities it creates.

2.6 Summary

Summarize the key insights from the literature review. Emphasize how the findings guide your project's direction or design choices.

3. System Analysis, Design and Implementation

3.1 Requirement Analysis

3.1.1 Functional Requirements

List and describe the specific functions that the system must perform. These requirements define what the system should do.

3.1.2 Non-functional Requirements

Describe performance, usability, reliability, and other quality attributes that define how the system should operate.

3.1.3 User Requirements Gathering

Summarize the data collected from potential users through surveys, interviews, or observations. Highlight user needs and preferences.

3.2 Feasibility Study

3.2.1 Technical Feasibility

Discuss whether the required technology, tools, and expertise are available to implement the project.

3.2.2 Economic Feasibility

Evaluate the cost-effectiveness of the project. Include estimations of cost, benefits, and return on investment.

3.2.3 Operational Feasibility

Assess whether the system will function as intended in the real-world environment and if users can adapt to it.

3.3 System Architecture

Describe the overall structure of the system, including major components, interactions, and deployment design.

3.4 Database Design

Provide the database design, including Entity-Relationship (ER) diagrams, schema structure, and relationships between entities.

3.5 UML Diagrams

3.5.1 Use Case Diagram

Illustrate how users (actors) interact with the system and identify major system functionalities.

3.5.2 Activity Diagram / Process Flow Diagram

Show the workflow or processes in the system through activity diagrams or process flow charts.

3.5.3 Sequence Diagram

Depict the sequence of interactions between system components or actors over time.

3.5.4 Class Diagram (if applicable)

Provide the class diagram showing system classes, attributes, methods, and relationships among them.

3.5.5 Data Flow Diagram (DFD) / Flowchart

Present the data flow diagrams or flowcharts to show how data moves through the system.

3.6 User Interface Design

Include visual representations of the user interface such as sketches, wireframes, or Figma prototypes. Describe key screens and interactions.

3.7 Security and Privacy Considerations

Discuss how the system ensures data security, user privacy, and protection against unauthorized access or misuse.

3.8 Tools and Technologies

3.8.1 Development Environment and Tools

Describe the hardware and software environment used for system development. Include IDEs, servers, databases, version control tools, and any other development utilities.

3.8.2 Programming Languages and Frameworks Used

List and explain the programming languages, frameworks, and libraries utilized in building the system. Justify why each was chosen for the implementation.

3.8.3 Implementation Details by Module

Provide detailed explanations of how each system module or component was implemented. Include relevant algorithms, logic, or code structure descriptions.

3.8.4 Integration of Modules

Explain how different modules were integrated to form a complete and functioning system. Discuss data flow, communication, and inter-module dependencies.

3.8.5 API / External Library Integration (if any)

Describe any external APIs, SDKs, or third-party libraries integrated into the system. Mention their roles, configuration steps, and interaction with internal components.

3.9 Challenges Faced During Implementation

Discuss the technical and non-technical challenges encountered during implementation and how they were resolved or mitigated.

3.10 Engineering Considerations

Engineering considerations encompass a set of crucial factors that engineers must weigh during the design, development, and implementation of projects.

3.10.1 Societal Impacts of Engineering Solutions

The relationship between the engineer and society involves recognizing the societal impacts of engineering solutions, including considerations for safety, accessibility, and the overall well-being of communities. Engineers are increasingly mindful of their role in promoting social responsibility and addressing diverse needs.

3.10.2 Environment and Sustainability Considerations

Environment and sustainability considerations involve assessing and mitigating the environmental impact of engineering projects. This includes minimizing resource depletion, reducing pollution, and promoting sustainable practices to ensure long-term ecological balance. Sustainable engineering aims to create solutions that meet present needs without compromising the ability of future generations to meet their own needs.

3.10.3 Ethical Considerations

Ethics in engineering emphasizes the moral and professional responsibilities of engineers. Ethical considerations involve ensuring the safety, honesty, and integrity of engineering practices. Engineers are expected to prioritize the well-being of individuals and communities, uphold transparency, and adhere to ethical codes in their decision-making processes.

In summary, engineering considerations encompass a holistic approach that integrates the engineer's relationship with society, environmental sustainability, and ethical practices. Striking a balance among these factors is essential for creating responsible and enduring engineering solutions.

4. Testing and Results Analysis

4.1 Testing

4.1.1 Testing Strategies

Describe the testing methodologies and strategies adopted. Explain how each type of testing (unit, integration, system, acceptance, automation (Selenium, Automation)) ensures the system's correctness and reliability.

4.1.2 Test Cases and Results (Manual and Automation)

Provide sample test cases, expected outcomes, and actual results. Include both manual and automated test scenarios, highlighting success and failure cases.

4.1.3 Performance Testing

Discuss performance evaluation under different loads or stress conditions. Present metrics such as response time, throughput, and scalability.

4.1.4 Security Testing

Explain the methods used to identify and mitigate security vulnerabilities. Mention penetration tests, data protection measures, and access control validations.

4.1.5 Usability Testing

Describe usability testing procedures involving real users. Summarize their feedback on system design, ease of use, and interface intuitiveness.

4.1.6 Bug Tracking and Fixes

Outline the process for identifying, logging, and fixing bugs. Mention tools used for bug tracking and provide examples of resolved issues.

4.1.7 Functional Performance Review

Assess how effectively the system performs its intended functions. Compare outcomes against performance benchmarks or KPIs.

4.2 Final System Output Screenshots

Include screenshots of the final implemented system to demonstrate functionality and visual design.

4.3 Comparison with Initial Requirements

Evaluate the final system against the original requirements. Indicate which requirements were fully, partially, or not implemented.

4.4 Strengths and Weaknesses of the System

Discuss the major strengths that make the system effective and the weaknesses or limitations that still exist.

4.5 User Feedback Analysis

Summarize and analyze user feedback obtained after deployment or testing. Highlight common opinions, suggestions, and areas of improvement.

5. Conclusion and Future Work

5.1 Summary of Achievements

Summarize the key accomplishments of the project. Highlight what has been successfully developed, implemented, or validated.

5.2 Fulfillment of Project Objectives

Discuss how the project objectives outlined earlier have been achieved. Indicate which goals were fully met and provide justification for any partial achievements.

5.3 Limitations of the Developed System

Describe the existing constraints or shortcomings of the system that may have affected its performance, scalability, or usability.

5.4 Recommendations for Enhancement

Provide suggestions for improving the system's design, performance, or functionality based on findings, feedback, or testing outcomes.

5.5 Future Scope of the Project

Discuss potential areas for future research or system expansion. Highlight new features, technologies, or integrations that could be explored to extend the project's usefulness.

Reference

[1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Advances in neural information processing systems*, vol. 25, pp. 1097–1105, 2012.

Appendices (Optional)

Include in the appendices information that could not be included in the formal body of the report because it would disrupt the continuity of the discussion. Background materials, experimental data tables, and extra documentation should be placed in the appendix.

6. How To Use This Template

Document-preparing WYSIWYG applications like Microsoft Word are general typing tools that are not friendly to the needs of scientific writing. It is difficult to create scientific writing components, such as equations, figures, and citations using such generic applications. Although some tools and plugins can be used with those applications there is an aesthetic issue that can only be mitigated using a typesetting tool, such as LaTeX. Besides, indexing and referencing are inherently easy to produce which helps the authors to concentrate more on writing than formatting.

Concretely, there are a few good reasons why LaTeX should be used for academic writing.

1. LaTeX allows you to typeset your thesis/project report beautifully. It helps to structure the document, equations, citations, indexing, etc, and produce high-quality output.

2. Editing a LaTeX document is easy. A part of the document can be easily moved and rested in the document allowing greater flexibility with the document without disturbing the typesetting.

3. Document reformatting using LaTeX is super easy. Changing fonts, margins, citation styles, and rearranging can be done in almost no time.

4. The scientific publishing industry has built a stable infrastructure around LaTeX.

5. LaTeX is free. Besides, the strong presence of a large LaTeX community in WWW to help you in producing high-quality documents.

If you are new to LaTeX, you are highly encouraged to read and practice the following instructions carefully before commencing your thesis/project report writing.

6.1 Scenario 1: A Student Doing A Thesis

A student of the Department of Computer Science and Engineering at the Green University of Bangladesh named *Afsin Fairuz*, Student ID 243014007, is working alone to write her thesis under the supervision of Associate Professor Dr. Md. Mostafizur Rahman. The title of her thesis is "Blockchain-Based Food Distribution on the Planet Mars." *Afsin* has already conducted her research and got her data ready for writing her thesis. She plans to defend the thesis in the Fall of 2028. The Department of Computer Science and Engineering has declared that *Afsin* will defend her thesis in **Board 3** consisting of Prof. Dr. Md. Saiful Azad (Professor, CSE) as chair, Mr. Shadman Wadith (Lecturer, CSE) as Member, and Mr. Shamsur Rahman (Tech Lead, Tiger IT Ltd) as External Member. The department also declared that the thesis defense date is February 02, 2029.

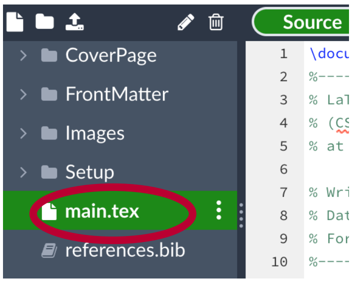

Figure 6.1: Gaining access to the *main.tex* file from the left side panel of Overleaf for doing initial configurations of this template.

Considering the scenario above, *Afsin* needs to do the initial configuration of this template. To do so, she has to click on the *main.tex* file as shown in Figure 6.1. Once *main.tex* is open, she should do the initial configurations as shown in the following two subsections.

6.1.1 Setting Thesis Particulars

Since *Afsin* is doing a thesis, she should declare it at the beginning. Thus in the template, she uses the `\def\ReportType{Thesis\xspace}` primitive and she comments out the other two `\def\ReportType{...\xspace}` primitives to make sure the type of her report is only Thesis. Moreover, the following primitives are to be filled according to the requirements.

- `\def\ReportTitle{...\xspace}`,
- `\def\Supervisor{...\xspace}`,
- `\def\SupervisorPosition{...\xspace}`,
- `\def\reportSubmissionDate{...}`, and
- `\def\reportSubmissionTerm{...}`.

After setting up the thesis particulars, the relevant part of the *main.tex* would look as follows.

```
%-----------------------------------------------------------
% Set your report/thesis particulars
%-----------------------------------------------------------
%
% Use one: Thesis/Project report
%\def\RoportType{Project report\xspace}
\def\RoportType{Thesis\xspace}
%
\def\ReportTitle{Blockchain-Based Food Distribution in the
Planet Mars\xspace}
%
\def\Supervisor{Dr. Md. Mostafizur Rahman\xspace}
\def\SupervisorPosition{Associate Professor\xspace}
%
%\def\reportSubmissionDate{\today}
\def\reportSubmissionDate{February 02, 2029}
\def\reportSubmissionTerm{Fall 2028}
```

6.1.2 Setting Author's Particulars

Based on the description provided in Subsection 6.1, there are no other members/authors in the group except *Afsin*. That means she is the first author of her Thesis. *Afsin* loves her parents, so she dedicates her thesis to her Mother *Mansura Akhter* and Father *Manzurul Haque*. She wants to write the following message on the "Dedication" page of the thesis.

To my loving mother
Mansura Akhter
and father
Manzurul Haque

So, *Afsin* would configure the following primitives as the requirements.

- `\def\numberOfAuthors{ }`
- `\def\firstAuthor{...\xspace}`
- `\def\firstAuthorID{...\xspace}`
- `\def\firstAuthorDedication{...}`

After inserting the author's particulars, the group members' particular part of the *main.tex* would look as follows.

```
%-----------------------------------------------------------
% Set your group members' particulars
%-----------------------------------------------------------
%
\def\numberOfAuthors{1} % write 1, 2, or 3 (depends on group)
%
\def\firstAuthor{Afsin Fairuz\xspace}
\def\firstAuthorID{243014007\xspace}
\def\firstAuthorDedication{To my loving mother\\ \textit{
Mansura Akhter} \\ and father \\ \textit{Manzurul Haque}}
```

Please note that *Afsin* does not have to fill up the remaining authors' information although there is a provision to input more authors' information. She can leave them as it will not create any problem in rendering her document.

Next, *Afsin* has to fill in the information regarding the defense board and its member information. Based on the information given, *Afsin* has to edit the following piece of code accordingly.

```
%-----------------------------------------------------------
% Set Board members' particulars
%-----------------------------------------------------------
%
\def\BoardNumber{3} % board to defende
%
\def\BoardChair{Prof. Dr. Md Saiful Azad\xspace}
\def\BoardChairPosition{Professor, Dept. of CSE\xspace}
%
\def\BoardMember{Mr. Shadman Wadith\xspace}
\def\BoardMemberPosition{Lecturer, Dept. of CSE\xspace}
%
\def\BoardExternalMember{Mr. Shamsur Rahman\xspace}
```

```
\def\BoardExternalMemberPosition{Tech Lead\xspace}
\def\BoardExternalMemberAffiliation{Tiger IT Ltd\xspace}
%
%-----------------------------------------------------------
```

Now *Afsin* is ready to recompile the document. In order to do that, she finds the 

button of Overleaf and click on it. Once recompilation is done, she finds that all the necessary changes are made automatically in the following pages.

1. Front page
2. Copyright page
3. Declaration page
4. Endorsement Certificate page
5. Certificate page
6. Acknowledgments page
7. Dedication page, and
8. Cover page

If you are doing a thesis alone, follow *Afsin* to configure the template according to the requirement. After configuration, you will be able to see the dynamically produced skeleton of your thesis too. Try it yourself! If you have done it successfully, Congratulations to you!

6.2 Scenario 2: A Group Doing A Project

Three students of the Department of Computer Science and Engineering at the Green University of Bangladesh formed a group to do a Project under the supervision of Lecturer *Jargis Ahmed*. Names of the students are *Safwan Sadid*, *Aiyan Faiyaz*, and *Aleena Ramin*, and their Student IDs are 243014008, 243014027 and 243014033 respectively. The group has decided the title of their project is “Design and Development of a Robot for Tree Plantation on Planet Mars.” The group members have already designed and implemented their project, and now they are ready to write the project report. Upon approval of their supervisor, they are planning to submit the report by the end of Fall 2028. The Department of Computer Science and Engineering has declared that the group will defend her project in **Board 3** consisting of Prof. Dr. Md. Saiful Azad (Professor, CSE) as chair, Mr. Shadman Wadith (Lecturer, CSE) as Member, and Mr. Shamsur Rahman (Tech Lead, Tiger IT Ltd) as External Member. The department also declared that the thesis defense date is February 02, 2029.

The group starts writing their project report doing the initial configuration of this template. To do so, first, they click on the *main.tex* file as shown in Figure 6.1. Once *main.tex* is open, they do the initial configurations of the template as shown in the following two subsections.

6.2.1 Setting Report Particulars

Since the group is doing a project, they use the `\def\ReportType{ Project\xspace}` primitive to make sure the type of their report is a project report. Further, the following primitives are filled accordingly.

- `\def\ReportTitle{...\xspace}`,
- `\def\Supervisor{...\xspace}`,
- `\def\SupervisorPosition{...\xspace}`,
- `\def\reportSubmissionDate{...}`, and
- `\def\reportSubmissionTerm{...}`.

After configuring the project report particulars, the relevant part of the *main.tex* would look as follows.

```
%------------------------------------------------------------
% Set your report/thesis particulars
%------------------------------------------------------------
%
% Use one: Thesis/Project report
\def\RoportType{Project report\xspace}
%\def\RoportType{Thesis\xspace}
%
\def\ReportTitle{Design and Development of a Robot for Tree
Plantation on Planet Mars\xspace}
%
```

```
\def\Supervisor{Jargis Ahmed\xspace}
\def\SupervisorPosition{Lecturer\xspace}
%
%\def\reportSubmissionDate{\today}
\def\reportSubmissionDate{February 02, 2029}
\def\reportSubmissionTerm{Fall 2028}
```

6.2.2 Setting Authors' Particulars

As described in Subsection 6.2, the group consists of three students who are working towards the successful completion of a project. So, all of them are considered as authors. Based on the contributions made to the project, they came up with an order in the authors' list.

Safwan, Aiyan, and Aleena – all of them decide to dedicate their project report to their favorite persons. Safwan wants to dedicate the work to his parents. The dedication message he wants to write is as follows.

To my beloved mother
Munjifa Hasan
and father
M Shohorab Hossain

Aiyan, on the other hand, loves his grandmother a lot so he decides to dedicate his work to his grandmother. The message goes as follows.

To my grandmother
Mahmuda Begum

Aleena has been influenced a lot by her mentor and her sister. So, she decides to dedicate the work to them. The messages she wants to include in the report are as follows.

I would like to dedicate my
work to my mentor
M. Tafazzal Hossain
and my sister
Samara Raniyah

Accordingly, the team configures the following primitives as the requirements.

- `\def\numberOfAuthors{ }`
- `\def\firstAuthor{...\xspace}`

- `\def\firstAuthorID{...\xspace}`
- `\def\firstAuthorDedication{...}`
- `\def\secondAuthor{...\xspace}`
- `\def\secondAuthorID{...\xspace}`
- `\def\secondAuthorDedication{...}`
- `\def\thirdAuthor{...\xspace}`
- `\def\thirdAuthorID{...\xspace}`
- `\def\thirdAuthorDedication{...}`

After inserting the authors' particulars, the group members' particular part of the *main.tex* would look as follows.

```
%------------------------------------------------------------
% Set your group members particulars
%------------------------------------------------------------
%
\def\numberOfAuthors{3} % write 1, 2 or 3 (depends on group)
%
\def\firstAuthor{Safwan Sadid\xspace}
\def\firstAuthorID{243014008\xspace}
\def\firstAuthorDedication{To my beloved mother\\ \textit{
Munjifa Hasan} \\ and father \\ \textit{M Shohorab Hossain}}
%
\def\secondAuthor{Aiyan Faiyaz\xspace}
\def\secondAuthorID{243014027\xspace}
\def\secondAuthorDedication{To my grandmother\\
\textit{Mahmuda Begum}}
%
\def\thirdAuthor{Aleena Ramin\xspace}
\def\thirdAuthorID{243014033\xspace}
\def\thirdAuthorDedication{I would like to dedicate my work to
my mentor\\ \textit{M. Tafazzal Hossain} \\ and my sister \\
\textit{Samara Raniyah}}
```

In the end, Board Members' particular part of the *main.tex* has to be filled up which would look as follows.

```
%------------------------------------------------------------
% Set Board members' particulars
%------------------------------------------------------------
%
\def\BoardNumber{3} % board to defende
%
\def\BoardChair{Prof. Dr. Md Saiful Azad\xspace}
\def\BoardChairPosition{Professor, Dept. of CSE\xspace}
%
\def\BoardMember{Mr. Shadman Wadith\xspace}
\def\BoardMemberPosition{Lecturer, Dept. of CSE\xspace}
%
\def\BoardExternalMember{Mr. Shamsur Rahman\xspace}
\def\BoardExternalMemberPosition{Tech Lead\xspace}
\def\BoardExternalMemberAffiliation{Tiger IT Ltd\xspace}
%
%------------------------------------------------------------
```

Now the team is ready to recompile the document. To do that, they click on the 

button to render the document. Once recompilation is done, they found the necessary changes according to the above configurations on the following pages.

1. Frontpage
2. Copyright page
3. Declaration page
4. Endorsement Page
5. Certificate page
6. Acknowledgments page
7. Dedication page, and
8. Cover page

If your team consists of three members and the team is doing a project then follow the above steps to complete the initial configurations. If you have done it successfully, to all three of you - Kudos!

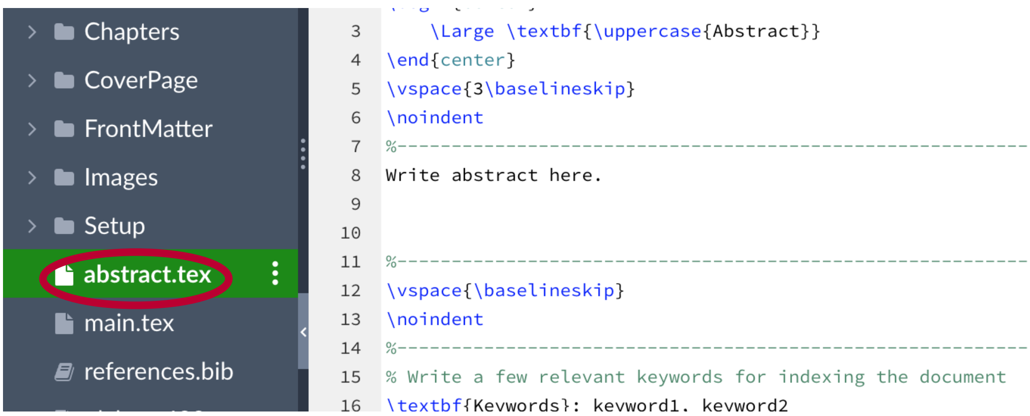

Figure 6.2: Write an Abstract of your work in the *abstract.tex* file accessible from the left side panel of Overleaf.

6.3 Abstract and Keywords

An abstract is a concise summary of an experiment or project written in usually a paragraph of roughly 250 words. An abstract should contain the information in the given order.

- the purpose of the study/project motivation;
- the basic design of the study/solving methods;
- major findings/challenges faced in implementing the project; and,
- a brief summary and conclusions.

In your report, you should write an abstract in the *abstract.tex* file. The location of the *abstract.tex* file is illustrated in Figure 6.2.

In the same file, there is a provision to include a few relevant keywords for indexing purposes.

6.4 Adding a New Chapter

Now, the template is ready for you to add chapters containing the text that you want to put in your report. This template already contains five chapters which are stored in the **Chapters** folder. You are free to edit all the chapters according to your needs. Additionally, if you need more chapters to cater to your needs, first, follow the instructions given in Figure 6.3. Then, from the **Add Files** dialogue box give a file name with *.tex* extension. Finally, click on the **create** button to confirm your intention.

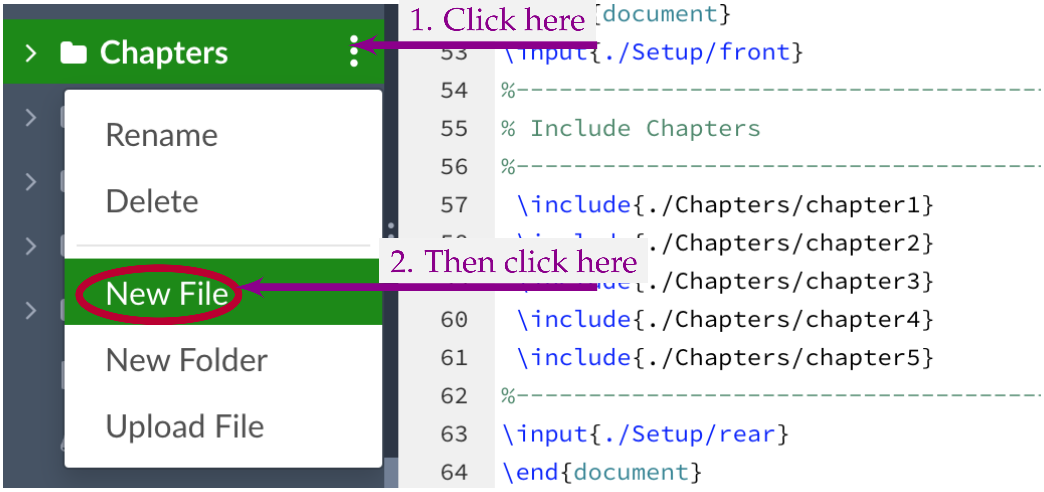

Figure 6.3: Create a new chapter from the left side panel of Overleaf of this template.

6.4.1 Customizing Chapter Title

The newly created chapter is to be prepared for use next. To customize the chapter title, you need to use `\chapter{}` command. For example, the newly created chapter title is *Literature Review*. In such case, you should use the following command at the beginning of the newly created chapter file.

```
\chapter{Literature Review}
```

The above command adds the newly created chapter. Now you are ready to add any content in the the chapter as required.

6.4.2 Including a Chapter in Your Report

Upon completing the chapter file creation and customizing the title, you can call it from *main.tex*. The primitive to be used in that case is `\include{./Chapters/...}`. Please replace ... with the chapter name that you created just a while ago.

```latex
%--------------------------------------------------------------
% Include Chapters
%--------------------------------------------------------------
 \include{./Chapters/chapter1}
 \include{./Chapters/chapter2}
 \include{./Chapters/chapter3}
 \include{./Chapters/chapter4}
 \include{./Chapters/chapter5}
```

Please note that you are free to rearrange the order of the chapters without considering their file name.

6.5 Adding Equations

It is almost impossible to write a thesis or a technical report in an engineering discipline without equations. These equations contain mathematical functions, symbols, parameters, numbers, units, or any combination of them. In LaTeX adding equations is typeset-based and easy to use. For example typeset `$\frac{a}{b}$` produces $\frac{a}{b}$, a simple typeset `$\bar{x} = \frac{\sum_{i = 1}^n x_i}{n}$` produces an elegant inline equation $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$, and a simple and easy looking typeset `$\cos (2\theta) = \cos^2 \theta - \sin^2 \theta$` produces an inline equation $\cos(2\theta) = \cos^2\theta - \sin^2\theta$.

Numbered equations are used and often required in scientific writing. To produce a numbered equation, you need to use an `equation` environment. The `equation` environment automatically numbers your equation which can be referred from the document. A simple example of generating a numbered equation using `equation` environment is as follows.

```latex
\begin{equation}
  x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
\end{equation}
```

The above set of commands adds the following numbered equation to your document.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{6.1}$$

You can also use the `\label` and `\ref` commands to label and reference equations, respectively. The following equation shows an example of how to use `\label` and `\ref` commands in context.

```
\begin{equation} \label{eq:2}
  E = mc^2
\end{equation}

The Eq. \ref{eq:2} defines the energy $E$ of a particle in its
rest frame as the product of mass $m$ with the speed of light
squared $c^2$.
```

The above piece of code produces the following output.

$$E = mc^2 \tag{6.2}$$

The Eq. 6.2 defines the energy E of a particle in its rest frame as the product of mass m with the speed of light squared c^2.

6.6 Adding a Section

While writing a scientific document, it is necessary and often important to structure the content of the document into fundamental logic units. To achieve this, LaTeX provides a command to generate section headings and number them automatically. The commands to create section headings are simple.

```
\section{...}
```

The `\section{}` command is numbered and it appears in the table of contents of the document.

6.6.1 Adding a Subsection

You can add a subsection like this one using `\subsection{}` command and works in a similar manner as `\section{}` does.

6.6.1.1 Adding a Sub-subsection

You can add a subsubsection like this one using `\subsubsection{}` command. Sometimes you need to structure your report to a finer level and `\subsubsection{}` the command helps you to achieve that intention.

6.7 Adding a Figure

"A picture is worth a thousand words" – a picture conveys information more effectively than words. A picture has the power of telling a story. sometimes complex and multiple ideas can be depicted using an image that would otherwise be difficult to convey by a verbal description. The images, figures, graphs, and charts published within research articles are often the most important information in the paper. Photos can sum up key findings in a single image.

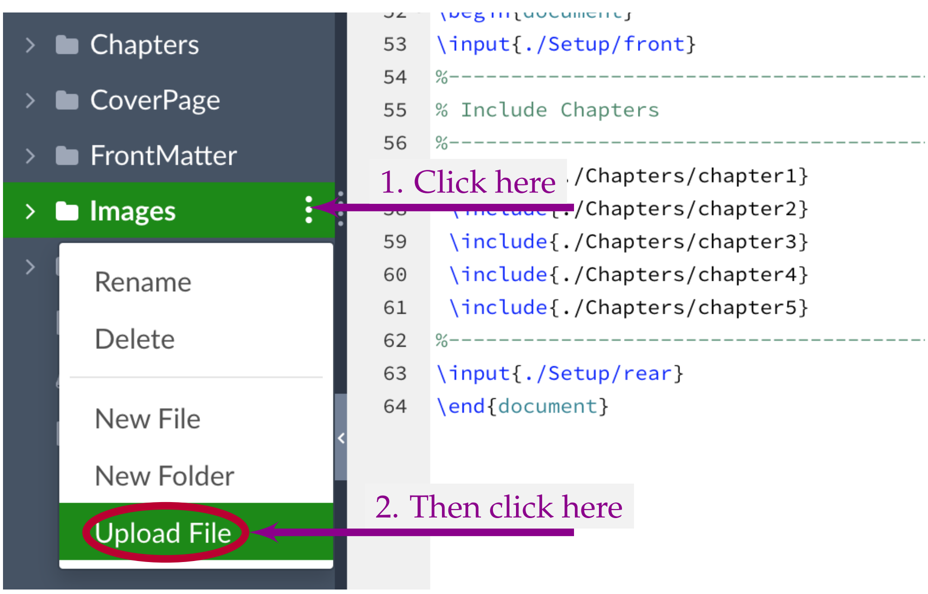

Figure 6.4: Adding a new figure in the Images folder from the left side panel of the Overleaf of this template.

There are many important reasons to use images for scientific and academic writing. However, most students do not realize it which results in losing the chance to make writing more eye-catching and interesting. This happens due to a lack of practice and lack of knowledge of academic writing. Additionally, sometimes students use images, graphics, and charts but they don't know how to include pictures in a research paper correctly with proper citing.

In your report, if you want to add a figure properly then the first and foremost important task is to import the figure in the "Images" folder of this project. To do that, please follow the instructions given in Fig. 6.4. Then use the *Add Files* dialogue box to complete uploading the image. When the figure is uploaded, you are now ready to use the image in your report.

Then add the figure to the desired location of your report using the following commands (assuming that the image you would like to add is Mars.jpg).

```
\begin{figure}[ht]
    \centering
    \includegraphics[width=0.40\textwidth]{Images/Mars.jpg}
    \caption{Picture of the Planet Mars in natural color.}
    \label{fig:mars}
\end{figure}
```

In the above code, a figure is added having the caption "Pictured of the Planet Mars in natural color.". Additionally, you should add a `\label{fig:mars}` to the figure by which you can refer the figure (e.g. `\ref{fig:mars}`) from the body of the text. The command `\centering` is used to center-align the figure.

You can also use the `\label` and `\ref` commands to label and reference figures, respectively. The following listing shows an example of how to use `\label` and `\ref` commands in context.

```
\begin{figure}[ht]
  \centering
  \includegraphics[width=0.40\textwidth]{Images/Einstein.png}
  \caption{Einstein's official portrait after receiving the
  1921 Nobel Prize in Physics.}
  \label{fig:einstein}
\end{figure}

In 1920, Albert Einstein became a Foreign Member of the Royal
Netherlands Academy of Arts and Sciences. In 1922, he was
awarded the 1921 Nobel Prize in Physics "for his services to
Theoretical Physics, and especially for his discovery of the
law of the photoelectric effect". Figure \ref{fig:einstein}
shows his official portrait after receiving the prize.
```

The above code produces the Figure 6.5 and the text below. Please have a careful look how `\lebel{}` is used within `figure` environment and how the figure is cited from the text using `\ref{}` command.

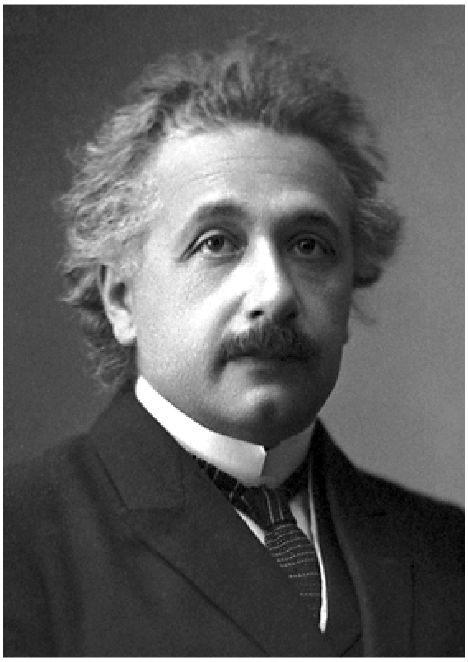

Figure 6.5: Einstein's official portrait after receiving the 1921 Nobel Prize in Physics.

In 1920, Albert Einstein became a Foreign Member of the Royal Netherlands Academy of Arts and Sciences. In 1922, he was awarded the 1921 Nobel Prize in Physics "for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect". Figure 6.5 shows his official portrait after receiving the prize.

6.8 Adding a Table

Tables are an integral part of scientific writing. A table often used to summarize your findings clearly and neatly, and it allows the reader to understand the results and importance of the results. When you plan to include a table in your report, you should think carefully to present your data so that the readers understand it easily. Tables have several elements, including the caption, column titles, and body.

- Caption: Tables should have a clear caption with sufficient amount of detail to understand it standalone.
- Column Titles: Column titles are the headline of your data. A good set of column titles allow the reader to understand the table quickly.

- Body: This is the part of the table where data is presented.

A table can be added in your documents by using a `table` environment. The following code of a `table` environment produces a table shown in Table 6.1.

```
\begin{table}[ht]
\centering
\caption{Data collected from four students.}
\begin{tabular}{l c c c}
\toprule
Name    & Weight (lb) & Height (in) & Gender \\ \toprule
Alice   & 133         & 65          & F      \\ \midrule
Bob     & 160         & 72          & M      \\ \midrule
Charlie & 152         & 70          & M      \\ \midrule
Diana   & 120         & 60          & F      \\ \bottomrule
\end{tabular}
\label{tab:1}
\end{table}
```

Table 6.1: Data collected from four students.

Name	Weight (lb)	Height (in)	Gender
Alice	133	65	F
Bob	160	72	M
Charlie	152	70	M
Diana	120	60	F

6.9 Citing Articles

In academic writing, you need to read many scientific articles, reports or even authentic web documents to know the state of the art of your field of research. While preparing your report, you may need to use the information provided in those documents. If you consider doing so, you are allowed to do it giving credits to the original authors via citations. A citation is a reference to the source article of information that you used in your research. Any time you directly quote, paraphrase or summarize the essential elements of others idea in your work, you must cite the source. While directly quoting sentences from a published work, it should be surrounded by quotations marks with proper citation. Paraphrasing and summarizing or the published work are also allowed but you should cite the source properly. To cite an article, you may copy the BibTeX for the corresponding article from Google Scholar or any digital library. Please follow the steps provided in Figure 6.6 to get the BibTeX of an article that you are looking for using Google Scholar.

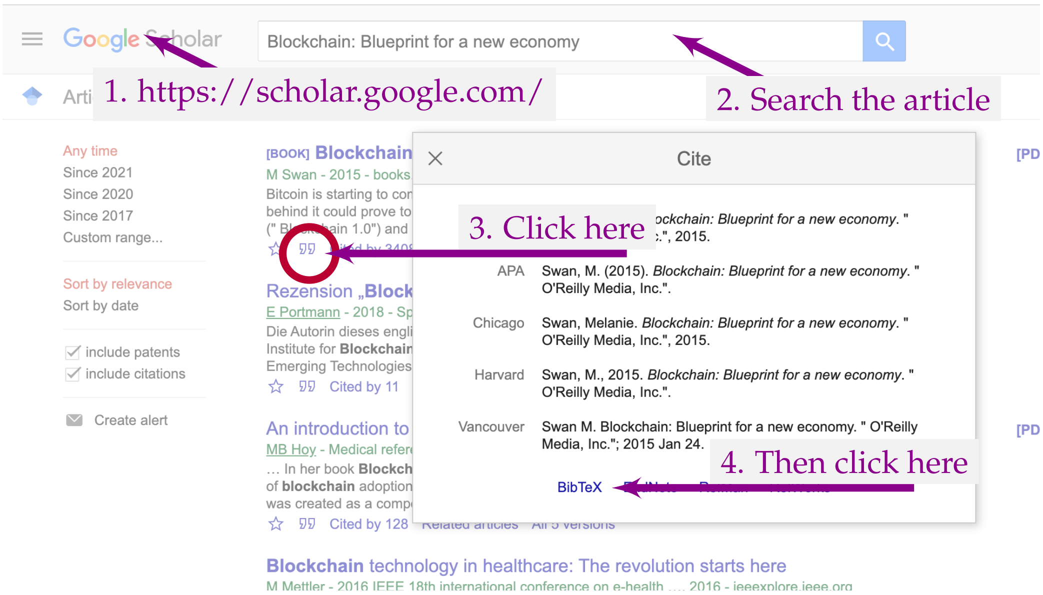

Figure 6.6: BibTeX is readily available in Google Scholar.

A typical BibTeX of an article looks as follows:

```
@article{krizhevsky2012imagenet,
  title={Imagenet classification with deep convolutional
  neural networks},
  author={Krizhevsky, Alex and Sutskever, Ilya and Hinton,
  Geoffrey E},
  journal={Advances in neural information processing systems},
  volume={25},
  pages={1097--1105},
  year={2012}
}
```

Get a required BibTeX and paste that in the *references.bib* file of this template. Please refer to Figure 6.7 to locate *references.bib* file. You should take a note of the key to use in your report. In the above example `krizhevsky2012imagenet` is the key.

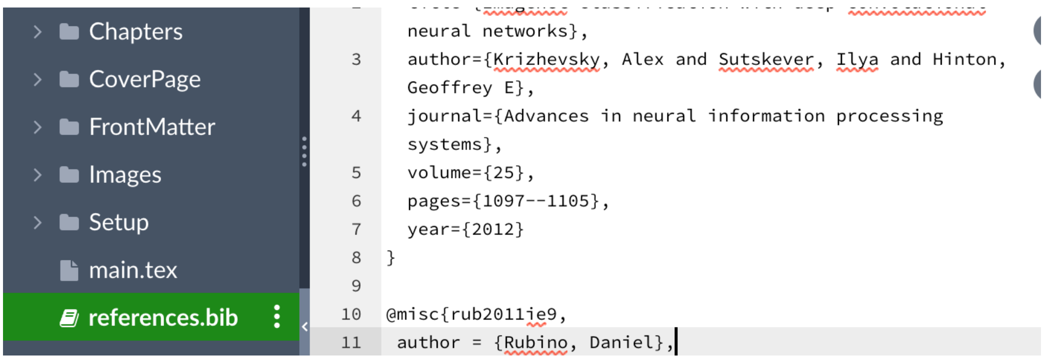

Figure 6.7: Adding a new BibTeX entry in reference.bib from the left side panel of Overleaf of this template.

Next, go to the desired location in your report to insert the reference. To cite this article, please write a command as follows:

```
Convolutional Neural Network revived with the success of
\cite{krizhevsky2012imagenet} in ImageNet competition.
```

Next time you recompile, reference will be created as follows:

> Convolutional Neural Network revived with the success of [1] in ImageNet competition.

Now, please go to the Bibliography section (you may jump to there by clicking on the number in green color as well) of your report where you will find bibliographic detail of your referred article.

6.10 Summary

LaTeX allows you to produce your document with a great level of flexibility without compromising the quality of the output. This chapter has described how to use this template with two scenarios. However, this template is not restricted to use only for those two cases. The scenarios provided here only for illustration purpose and for better understanding of the students. You are allowed to use it in any combination of a group and the report type. For any support please send an email to muhammad.hasan@cse.green.edu.bd

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by

First Author
243014001

Project for the degree of

Bachelor of Science in Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GREEN UNIVERSITY OF BANGLADESH

FALL 2025